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I TA0518 - computer vision

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1. A dark image lacks detail.

* The best grey level transformation is Log Transformation.

* It increases the brightness of dark pixels while keeps bright pixels almost unchanged.

* This makes hidden details in dark region visible.

* Result: The image becomes brighter and details in low intensity areas are clearly visible.

2. An image has Poor contrast.

* The suitable technique is Histogram Equalization.

* This improves the contrast of the image.

* Application: Compute the histogram and equalize the intensity values.

* Result: The image looks clearer and objects become easier.

3. A noisy image contains random intensity variations.

Ans

- * The best Smoothing filter is the mean filter.
- * It replaces each pixel with the average of its neighbours pixels.
- * This reduced random noise.
- * Result: The image becomes smoother, but very fine details may be slightly

blurred.

4.) An object clearly brighter than the background.

=> Ans * The suitable Segmentation technique is Thresholding.

- * A Threshold value separates the object from the background.
- * Result: The object is extracted quickly and accurately.

5.) An image contains Salt and Pepper noise.

- * The best filter is the median filter.
- * It replaces each pixel with the median value of its neighbouring pixels.
- * It removes Salt and Pepper noise.
- Result: Noise is removed.